

Software Requirement Specification

Purchase Requisition Management System (PRMS) with AI-Powered Item Intelligence

Overview & Workflow

Project Introduction:

The Purchase Requisition Management System (PRMS) automates the complete purchase requisition process from request creation to procurement completion.

Objectives:

- Digitize procurement workflow
- Improve transparency and accountability
- Reduce approval turnaround time
- Enable real-time tracking
- Provide AI-assisted item intelligence

User Roles:

- Requestor
- Department HOD
- Sourcing Department HOD
- CEO / Administrative Director
- Purchase Department
- Store / Materials Department
- Administrator

Workflow:

Requestor → Department HOD → Sourcing Department HOD → CEO

At each approval stage:

- Approve & Forward
- Return with Remarks
- Reject with Remarks

Purchase Request Module

Request Creation:

Auto Generated:

- Request Number
- Request Date

Auto Fetched:

- Requesting Department

- Cost Center
- Requested By
- Department Name

User Inputs:

- Priority (Routine / Urgent / Emergency)
- Justification / Purpose

Item Details:

- Item Name
- Item Category
- Brand
- Specification
- Unit of Measure
- Quantity Requested
- Estimated Cost
- Preferred Vendor
- Remarks

Supports multiple items within a single request.

Approval Workflow & Tracking

Approval Stages:

- Department HOD Approval
- Sourcing Department HOD Review
- CEO Approval

Actions:

- Approve & Forward
- Return with Remarks
- Reject with Remarks

AI-Powered Item Intelligence

The system shall include an integrated AI-powered item search feature that allows users to obtain detailed information about requested items directly within the application without visiting external websites.

Users can search using:

- Item Name
- Brand
- Model Number
- Keywords

AI Features:

- Product Description
- Specification Suggestions
- Category Identification
- Department Mapping
- Estimated Market Information
- Alternative Products
- Auto-fill Item Details

Phase 1:

Uses affordable general-purpose LLMs to provide reliable procurement-related information at low operational cost.

Phase 2 (Optional Premium Upgrade):

Supports Gemini, ChatGPT, and Claude for:

- Real-time Web Search
- Manufacturer Websites
- Product Datasheets
- Product Brochures
- External Product Information

Reports,& Future Scope

Reports:

- Department-wise Requests
- Pending Requests
- Approved Requests
- Rejected Requests
- Procurement Summary
- AI Usage Report